

# Genetic Programming Hyper-Heuristics for the Job Shop Scheduling Problem with Interval Durations: Interpretable Rules that Generalize Across Instance Sizes

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## Abstract

Dispatching rules remain the method of choice for scheduling under tight time budgets, but hand-crafted rules leave substantial performance unexploited, and the automated design of rules has scarcely been explored for scheduling problems with non-deterministic processing times. We present a genetic programming (GP) hyper-heuristic for the job shop scheduling problem with interval durations (IJSP), in which operation durations are known only to lie within an interval and schedules are evaluated by interval arithmetic. The GP evolves priority expressions over an *interval-aware* terminal set that exposes both the worst-case values and the widths of the uncertain quantities, so that reacting to uncertainty itself is expressible. On the 70 interval versions of Taillard’s benchmark, over 30 independent evolutions the resulting rules average a relative error of 18.99% ( $\pm 1.33$ ) with a single deterministic constructive pass, and the best evolved rule attains 17.71%—improving on the strongest deterministic constructive baseline (Giffler–Thompson with MWKR tie-breaking, 29.5%) on every single instance—and generalizes zero-shot across all seven instance size classes

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despite being evolved on only four training instances of a single size class ( $20 \times 15$ ). Applied without any re-evolution to a second benchmark of classical instances, the rules keep the same margin over the constructive baselines that they hold on the benchmark they were evolved for, and a GRASP-style randomized scheme closes much of the remaining distance to metaheuristics that search each instance separately. Evolved rules are compact, human-readable expressions, with the evolutionary hyperparameters configured automatically by irace. A controlled ablation over 30 evolutions per arm characterizes what interval awareness buys: the width terminals do not improve the expected makespan—the worst-case bounds carry that signal—but they yield significantly narrower makespan intervals, modestly under a makespan objective and substantially under one that rewards predictability, a trade-off the user can dial. We position the method against fixed rules, active schedule generators, and published metaheuristics.

*Keywords:* interval job shop scheduling, genetic programming, hyper-heuristics, dispatching rules, scheduling under uncertainty, robustness

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## 1. Introduction

Real manufacturing environments rarely know the exact duration of an operation in advance. The interval job shop scheduling problem (IJSP) models this uncertainty in a minimal, distribution-free way: each processing time is only assumed to lie within a known interval, and schedules are built and evaluated with interval arithmetic, so that the makespan of a solution is itself an interval and solutions are compared by an interval ranking. Population metaheuristics for the IJSP—artificial bee colony variants and tabu search among them—achieve strong results, at the cost of building and evaluating

large numbers of complete solutions for every instance [2, 3].

At the opposite end of the computational spectrum are *dispatching rules*: priority functions that build a schedule in a single constructive pass by repeatedly selecting the next operation to start. Their appeal is threefold—speed, interpretability, and trivial deployment—but hand-crafted rules (SPT, MOR, MWKR and relatives) leave a large quality gap with respect to search-based methods, as the experiments reported later in this paper confirm.

For *deterministic* scheduling, a rich literature has shown that this gap can be narrowed substantially by evolving rules automatically with genetic programming (GP) [18, 19]. For scheduling under interval uncertainty, however, the automated design of dispatching rules remains, to the best of our knowledge, unexplored. Moving automated rule design to interval durations raises questions that do not arise in the deterministic case: which attributes a rule should read, when the quantities describing a candidate operation—its duration, its earliest start, the work remaining in its job—are intervals rather than numbers; how candidate operations should be prioritized, and the resulting schedules compared, given that intervals admit no natural total order; and whether the *magnitude* of the uncertainty carries information that is useful for dispatching, or whether it is the position of the intervals that drives the decision. Answering these questions empirically requires exposing uncertainty to the rule as first-class attributes, which is the starting point of this work.

This paper makes the following contributions:

1. A GP hyper-heuristic for the IJSP whose terminal set is *interval-aware*: alongside worst-case processing time, earliest start and work remaining, rules can read the *widths* of these quantities, so that uncertainty-

sensitive prioritization is expressible at all (Section 4).

2. An empirical study on the 70 interval Taillard instances showing that evolved rules (i) beat every deterministic constructive baseline by a wide margin at equal cost, (ii) generalize zero-shot across instance sizes—evolved on  $20 \times 15$ , best class  $50 \times 15$ —and (iii) transfer to a second benchmark of classical instances, keeping the same margin over the constructive baselines (Section 6).
3. An analysis of *why* the rules work: an anatomy of the evolved expressions (Section 7.1); an ablation that isolates the contribution of the interval-width terminals: they leave the expected makespan unchanged but yield significantly narrower makespan intervals, modestly under a makespan objective and substantially under an objective that rewards predictability (Section 7.2); and an executional-robustness study showing that the evolved rule yields schedules whose realised makespan deviates less from its prediction than the baselines’, increasingly so as uncertainty grows (Section 7.3).
4. An automatic configuration study (irace, 200 experiments) of the evolutionary hyperparameters, whose winning configuration is used throughout (Section 5.3).

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 formalizes the IJSP. Section 4 presents the GP hyper-heuristic, from both the hyper-heuristic and the evolutionary viewpoints. Section 5 details the benchmark, the training protocol and the irace configuration study. Section 6 reports the empirical results, and Section 7 analyses them: rule anatomy, the interval-awareness ablation, executional robustness and limitations. Section 8 concludes.

## 2. Related Work

### *2.1. Job shop scheduling under interval uncertainty*

Interval durations offer a distribution-free uncertainty model requiring only bounds, in contrast to stochastic and fuzzy approaches [27]: they are the natural choice when historical data are insufficient to estimate distributions and a decision maker can only supply a minimum and a maximum duration. Scheduling with interval processing times has been studied across machine environments—single machine [12], flow shop, and flexible job shop [14], among others surveyed in [13]—and also analytically, through measures of the problem uncertainty induced by the intervals and of the stability of optimal sequences [11].

For the job shop with makespan minimisation, which concerns us here, the literature is dominated by population-based metaheuristics. Lei’s population-based neighbourhood search [10], which ranks interval makespans by a possibility degree, was an early reference point; it was subsequently outperformed by a genetic algorithm with a systematic study of interval rankings [1] and by artificial bee colony variants [2, 3], which report the best published results for this problem. Related objectives have also been addressed under interval uncertainty, notably tardiness minimisation [15]. All of these are iterative search methods: they build and evaluate a population of complete solutions and refine it over many iterations for each instance solved. Fast constructive methods for the IJSP have received far less attention; dispatching rules with interval-aware lexicographic comparisons are the de facto baseline.

## 2.2. Dispatching rules for job shop scheduling

Priority dispatching rules are among the oldest and most widely deployed scheduling heuristics: at every decision point, a priority function ranks the operations that are ready to start and dispatches the best one, building a schedule in a single constructive pass. Classical rules score each ready operation by a single attribute, and we adopt their standard acronyms throughout this paper. *Shortest* and *longest processing time* (SPT, LPT) look at the duration of the operation itself. *Earliest starting time* (EST) prefers the operation that can begin soonest given the current state of its job and its machine. *Most work remaining* (MWKR) and *most operations remaining* (MOR) look instead at the job the operation belongs to, and favour the one with the largest amount of pending work, measured respectively in time units and in number of operations. The *critical ratio* (CR) weighs the work still to be done against the time available until a due date. Surveys catalogue more than a hundred such variants [7, 8]. A step above single-attribute rules, the algorithm of Giffler and Thompson (G&T) [4] restricts each choice to a machine’s conflict set, guaranteeing *active* schedules and markedly improving plain rules when a priority criterion is used to break ties inside that set; we write G&T-SPT and G&T-MWKR for the resulting combinations. Stronger constructive procedures exist for the *deterministic* job shop, notably the shifting bottleneck heuristic [5], which repeatedly solves one-machine subproblems with heads and tails by means of Carlier’s algorithm [6]. These fall outside the scope of the present work: their machinery rests on exact one-machine optimisation and critical-path arguments whose extension to interval arithmetic, where no natural total order is available, is a research contribution in itself and remains open. Decades of practice consolidated two lessons: rules are unmatched in speed, transparency and ease of deploy-

ment, but hand-crafting them plateaued—no fixed combination of attributes performs well across shop configurations and objectives [8]. That plateau is what motivated the *automated* design of rules.

### 2.3. Automated design of dispatching rules

Genetic programming [16, 17] evolves a population of expressions, usually represented as trees, by applying selection and variation operators directly on their structure. GP hyper-heuristics apply this machinery to *priority expressions* over scheduling attributes, and constitute the standard approach to automated rule design for production scheduling [18, 19]. Evolved rules have been shown to outperform hand-crafted ones across job shop variants, with active research on surrogate fitness, feature selection, and multi-objective formulations. Beyond single rules, a productive line combines several evolved rules into *ensembles* that jointly construct or vote on a solution [22], and recent work explores alternatives to evolutionary search for the same design task—systematic tree search over the space of expression trees [23]—reporting rules of comparable quality but smaller size and better readability. All of this work targets deterministic (or dynamic-stochastic) settings. Closest to our setting, Karunakaran et al. [20] evolve rules for dynamic job shops with uncertain processing times, exposing summary statistics of observed deviations (an exponential moving average) to the rule; uncertainty there is realized stochastically during simulation, and rules are evaluated on sampled scenarios. A recent survey of learned optimisation for the job shop [21], covering both GP and DRL constructive approaches, records no work on interval-valued processing times. To the best of our knowledge, evolving rules whose *inputs* are interval attributes—and whose fitness is computed by interval arithmetic rather than by sampling realizations—has not been

explored before.

Finally, deep reinforcement learning provides an alternative family of learned constructive methods for the JSP [25, 26]; GP rules and DRL policies are increasingly studied as competing learning paradigms for the same construction task, and controlled comparisons between them under common protocols have been identified as a gap in the recent literature [21]. Both families occupy the same computational class at deployment: a single constructive pass.

### 3. The Interval Job Shop Scheduling Problem

An IJSP instance consists of  $n$  jobs and  $m$  machines. Job  $j$  is a chain of  $m$  operations  $o_{j1} \prec o_{j2} \prec \dots \prec o_{jm}$ , each requiring a distinct machine  $\mu(o_{jk})$  exclusively and non-preemptively. The duration of operation  $o$  is an interval  $p_o = [\underline{p}_o, \bar{p}_o]$  with  $0 < \underline{p}_o \leq \bar{p}_o$ : the only assumption is that the realized duration lies within these bounds.

Schedule construction uses interval arithmetic. For intervals  $a = [\underline{a}, \bar{a}]$  and  $b = [\underline{b}, \bar{b}]$ , the two operations required by makespan minimisation are

$$a + b = [\underline{a} + \underline{b}, \bar{a} + \bar{b}], \quad (1)$$

$$\max(a, b) = [\max(\underline{a}, \underline{b}), \max(\bar{a}, \bar{b})], \quad (2)$$

both component-wise [1]. A solution determines a task processing order  $\pi$  (a permutation with repetition of jobs, decoded left to right). Following the notation of [1, 3], let  $PM_o(\pi)$  and  $PJ_o$  denote the predecessors of task  $o$  in its machine (according to  $\pi$ ) and in its job. The *semi-active* schedule [9] starts each task at

$$\mathbf{s}_o(\pi) = \max(\mathbf{c}_{PJ_o}(\pi), \mathbf{c}_{PM_o(\pi)}(\pi)), \quad \mathbf{c}_o(\pi) = \mathbf{s}_o(\pi) + \mathbf{p}_o, \quad (3)$$

so starting and completion times of all tasks are themselves intervals. The problem can be stated as

$$\min_R \mathbf{C}_{\max} \quad (4)$$

$$\text{s.t. } \mathbf{C}_{\max} = \max_{1 \leq j \leq n} \{\mathbf{c}_{o(j, m_j)}\}, \quad (5)$$

$$\underline{s}_{o(j, l)} \geq \underline{c}_{o(j, l-1)}, \quad \bar{s}_{o(j, l)} \geq \bar{c}_{o(j, l-1)}, \quad 1 \leq l \leq m_j, \quad 1 \leq j \leq n, \quad (6)$$

$$\underline{s}_o \geq \underline{c}_{o'} \vee \underline{s}_{o'} \geq \underline{c}_o, \quad \bar{s}_o \geq \bar{c}_{o'} \vee \bar{s}_{o'} \geq \bar{c}_o, \quad \forall o \neq o' : \mu(o) = \mu(o'), \quad (7)$$

where Eq. (5) defines the interval makespan as the component-wise maximum of the job completion times, Eq. (6) enforces precedence within each job and Eq. (7) forbids overlaps on each machine, in both interval components [1, 3]. The minimum in (4) is taken with respect to an interval ranking  $R$ , since intervals admit no natural total order. We rank schedules lexicographically, comparing upper bounds first and using the lower bound only to break ties:

$$\mathbf{a} \preceq \mathbf{b} \iff \bar{a} < \bar{b} \vee (\bar{a} = \bar{b} \wedge \underline{a} \leq \underline{b}), \quad (8)$$

one of the standard rankings of the IJSP literature and the one found in [1] to yield the most robust schedules. All methods in this paper—the GP rules and every baseline—share one implementation of this decoder and evaluator, so reported differences cannot stem from evaluator discrepancies.

Performance is reported as the relative error of the expected makespan with respect to a crisp reference bound  $LB$  of each instance,

$$\text{RE} = \frac{E[\mathbf{C}_{\max}] - LB}{LB} \times 100, \quad E[\mathbf{C}_{\max}] = \frac{\underline{C}_{\max} + \bar{C}_{\max}}{2}, \quad (9)$$

where  $E[\mathbf{C}_{\max}]$  is the expected value of the interval makespan under the uniform distribution on  $[\underline{C}_{\max}, \bar{C}_{\max}]$ , and  $LB$  is a bound of the underlying

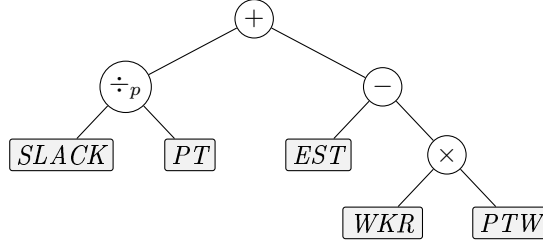


Figure 1: An individual is a priority expression tree; leaves are interval-aware attributes of the candidate operation (here the illustrative rule  $SLACK/PT + EST - WKR \cdot PTW$ ). At each decision point the tree is evaluated on every eligible operation and the lowest-priority one is dispatched.

deterministic instance—Taillard’s bounds for the Taillard-derived instances, the best known ones [2] for the classical set. It serves as an indicative common reference, not as a bound for the interval problem, for which none is available.

A schedule is eventually *executed* under one realization of the durations, and a second quantity of interest is how well its a-priori interval predicts that execution. Following [1], the  $\bar{\varepsilon}$  robustness of a schedule with a fixed processing order is the average relative deviation of its executed makespan from the predicted expected value, over  $K$  realizations of the durations:

$$\bar{\varepsilon} = \frac{1}{K} \sum_{k=1}^K \frac{|C_{\max}^{ex,k} - E[\mathbf{C}_{\max}]|}{E[\mathbf{C}_{\max}]}, \quad (10)$$

where  $C_{\max}^{ex,k}$  is the crisp makespan obtained by executing that same order when the durations turn out to take the values of realization  $k$ . A low  $\bar{\varepsilon}$  therefore means that what was promised before execution is close to what is obtained after it.

Table 1: Interval-aware terminal set, with exact definitions. For an eligible operation  $o$  of job  $j$  on machine  $m$ :  $p_o = [\underline{p}_o, \bar{p}_o]$  is its duration interval;  $C_j$  and  $C_m$  are the (interval) completion times of the last scheduled operation of job  $j$  and machine  $m$ ;  $\mathcal{R}_o$  is the set of operations of  $j$  after  $o$ ;  $\mathcal{E}$  is the current eligible set. Terminals are evaluated on *raw* (unnormalized) values.

| Terminal | Definition                                                                                      | Role                            |
|----------|-------------------------------------------------------------------------------------------------|---------------------------------|
| PT       | $\bar{p}_o$                                                                                     | worst-case processing time      |
| PTW      | $\bar{p}_o - \underline{p}_o$                                                                   | duration uncertainty            |
| EST      | $\bar{e}_o = \max(\bar{C}_j, \bar{C}_m)$                                                        | worst-case earliest start       |
| ESTW     | $\bar{e}_o - \underline{e}_o$ , with $\underline{e}_o = \max(\underline{C}_j, \underline{C}_m)$ | start uncertainty               |
| WKR      | $\sum_{q \in \mathcal{R}_o} \bar{p}_q$                                                          | worst-case work remaining       |
| WKRW     | $\sum_{q \in \mathcal{R}_o} (\bar{p}_q - \underline{p}_q)$                                      | remaining-work uncertainty      |
| NOR      | $ \mathcal{R}_o $                                                                               | operations remaining            |
| SLACK    | $\bar{e}_o - \min_{o' \in \mathcal{E}} \bar{e}_{o'}$                                            | start slack within eligible set |
| ONE      | 1                                                                                               | constant                        |

#### 4. GP Hyper-Heuristic for the IJSP

We describe the method from two complementary viewpoints: as a *hyper-heuristic*—what an individual is and how it turns an IJSP instance into a schedule (Section 4.1)—and as an *evolutionary algorithm*—how the population of rules is initialized, varied and selected (Section 4.2).

##### 4.1. Hyper-heuristic view: from expression trees to schedules

An individual is a priority expression tree (Figure 1). Leaves are drawn from the interval-aware terminal set of Table 1, and internal nodes from the function set

$$\mathcal{F} = \{ +, -, \times, \div_p, \min, \max, \text{neg} \}, \quad (11)$$

where  $\text{neg}$  is the unary sign change and  $\div_p$  denotes protected division, defined as  $a \div_p b = a/b$  if  $|b| > 10^{-9}$  and  $a$  otherwise. The rule is deployed inside a semi-active schedule generation scheme (SGS): starting from the empty schedule, at each of the  $N$  decision points the set  $\mathcal{E}$  of eligible operations (those whose job predecessor is already scheduled) is formed, the tree is evaluated on the attributes of every  $o \in \mathcal{E}$ , and the operation with *lowest* priority value is appended at its earliest feasible start on its machine, Eq. (3).

Because the lowest value is dispatched, a criterion that seeks to *maximise* an attribute must be expressed through the unary  $\text{neg}$  operator. Classic rules are thus points of this search space: shortest processing time is the single-node tree  $PT$ , whereas most work remaining is  $\text{neg}(WKR)$ . We inject both as seed individuals in the initial population, so that the search departs from a population that already contains well-known rules rather than from purely random expressions.

Priority ties at dispatch time are broken by eligible set order (lowest job index first), which is deterministic; hence a rule maps each instance to exactly one schedule.

#### 4.2. Evolutionary components

The initial population is built with the *ramped half-and-half* method, introduced by Koza [16] and still the standard choice for tree-based GP [17], which combines two tree-generation procedures over a range of depths. The *full* method expands every branch with function nodes until the prescribed maximum depth is reached, placing terminals only at that level, and thus produces complete trees whose leaves all lie at the same depth, whereas the *grow* method may close a branch earlier by placing a terminal—with probability 0.3 at each node in our implementation—and thus produces trees

of irregular shape and variable size. In both cases the function at an internal node is drawn uniformly from  $\mathcal{F}$  and a leaf uniformly from the terminal set, the arity of the chosen function determining the number of branches to expand. Each individual draws its maximum depth uniformly from  $\{2, 3, 4\}$  and is generated by either method with equal probability. The last two members of the population are not generated at random: they are the two seed rules of Section 4.1, inserted directly. Every individual is evaluated by the mean RE of its deterministic constructive pass over the four training instances, computed with the same environment, decoder and evaluator used by all baselines.

Each generation then proceeds as follows. Parents are chosen by tournament, drawing  $k$  individuals uniformly without replacement from the current population and returning the fittest of them. With probability  $p_c$  two parents are recombined by *subtree crossover* [16, 17], the standard recombination operator for tree-based GP. The offspring is a copy of the first parent in which one subtree has been replaced by a subtree taken from the second: a node is drawn uniformly at random in each parent, leaves included, and the subtree rooted at the node chosen in the first parent is substituted by the subtree rooted at the node chosen in the second. In its classical formulation the operator exchanges both subtrees and returns two offspring; we keep a single one per recombination. With the complementary probability  $1 - p_c$  a single parent undergoes *subtree mutation* [16, 17]: a node is again drawn uniformly at random, and the subtree rooted at it is discarded and replaced by a freshly generated random tree, built with the grow method and maximum depth 3 over the full primitive set. Offspring whose size exceeds a cap of  $c$  nodes receive infinite fitness and are thereby excluded from selection. Note that the depth bounds above apply only to the trees created from scratch, at initial-

Table 2: Evolution parameters: the configuration selected by irace (Section 5.3), used for every experiment reported in this paper.

|                            |                                                    |
|----------------------------|----------------------------------------------------|
| Population / generations   | 100 / 50                                           |
| Initialization             | ramped half-and-half (depths 2–4) + 2 seeded rules |
| Selection                  | tournament, size $k = 7$                           |
| Crossover / mutation prob. | $p_c = 0.77$ / $0.23$ (subtree, depth $\leq 3$ )   |
| Elitism                    | $e = 2$                                            |
| Tree-size cap              | $c = 30$ nodes (violations: fitness $\infty$ )     |
| Fitness                    | mean RE, deterministic rollout, TA11–TA14          |
| Seeds                      | 1–30 (RNG of evolution and environment)            |

ization and inside mutation: the depth of an individual is never constrained during the run, so recombination routinely produces expressions deeper than those of the initial population, the number of nodes being the only quantity that is bounded.

Replacement is fully generational, the offspring becoming the next population, except for the  $e$  best individuals of the current generation, which are copied into it unchanged. The process stops after a fixed number of generations and returns the best rule found. Algorithm 1 summarizes the generational loop and Table 2 lists the parameter values.

#### 4.3. Randomized multi-start extension

A deterministic rule produces one schedule. For matched comparisons with sampling-based methods we wrap the evolved rule in GRASP-style randomization: with probability  $\varepsilon=0.1$  the dispatched operation is drawn uniformly from the eligible set, otherwise the rule decides; sample 0 remains deterministic. Best-of- $N$  then reports the lexicographically best schedule. This uniform perturbation of a single rule is the simplest member of a broader

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**Algorithm 1** GP hyper-heuristic for the IJSP

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1:  $P \leftarrow \text{ramped half-and-half}(\text{pop} - 2) \cup \{\text{SPT}, \text{MWKR}\}$ 
2:  $\text{evaluate}(P)$   $\triangleright$  mean RE of one constructive pass on the training
   instances
3: for  $g = 1$  to  $\text{gens}$  do
4:    $P' \leftarrow \text{elite}(P, e)$   $\triangleright$  elitism
5:   while  $|P'| < \text{pop}$  do
6:     if  $\text{rand}() < p_c$  then
7:        $child \leftarrow \text{subtreeCrossover}(\text{tournament}(P, k),$ 
    $\text{tournament}(P, k))$ 
8:     else
9:        $child \leftarrow \text{subtreeMutation}(\text{tournament}(P, k))$ 
10:    end if
11:     $P' \leftarrow P' \cup \{child\}$   $\triangleright$  fitness  $\infty$  if size  $> c$ 
12:  end while
13:   $P \leftarrow \text{evaluate}(P')$ 
14: end for
15: return best rule in  $P$ 
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family of randomized rule-based builders; a richer alternative, orthogonal to our contribution, swaps among a *set* of evolved rules during construction [24].

## 5. Experimental Setup

### 5.1. Benchmark instances

We use two benchmarks, both taken from the IJSP literature [1, 2, 3]: the 70 interval versions of Taillard’s instances (TA1–TA70, seven size classes from  $15 \times 15$  to  $50 \times 20$ ), and the 12 classical instances on which the published

IJSP metaheuristics report results (FT10, FT20, six instances of the La family and ABZ7–9, of sizes  $10 \times 10$  to  $20 \times 15$ ). Both were obtained from their crisp counterparts with the generation scheme introduced in [1]: for every operation a value  $\delta$  is drawn uniformly at random in  $[0, 0.15p]$ , where  $p$  is its original duration, and that duration becomes the integer interval  $[p - \delta, p + \delta]$ , machine routes being unchanged. The interval is symmetric around  $p$ , so the crisp instance is exactly the midpoint scenario of its interval version. Since rules are evolved on Taillard instances only and applied to the classical set without any re-evolution, the latter doubles as a cross-family generalization test (Section 6.4). Both sets are released with this paper (see Data availability).

### 5.2. Training protocol, data split and baselines

Rules are evolved on TA11–TA14 ( $20 \times 15$ ), and TA15–TA20 serve as development set: they never contribute to fitness, but design and configuration decisions (including the irace study of Section 5.3) were selected on their performance, so we flag them as subject to development reuse. The remaining 60 Taillard instances and the 12 classical instances are fully unseen by every design decision and constitute the honest generalization test.

Each evolution runs a population of 100 rules for 50 generations and returns the best rule of its final population. Thirty independent evolutions, differing only in their random seed, therefore yield thirty rules; these are evaluated on the 70 benchmark instances, and the one with the lowest mean RE is the *featured rule*.

### 5.3. Hyperparameter configuration with irace

The four qualitative knobs of the evolution—tournament size, crossover probability, tree-size cap and elitism—were configured with irace over a bud-

Table 3: Hyperparameter space explored by irace (200 experiments; cost = development-set RE of the evolved rule) and the winning configuration.

| Parameter             | Range / values   | irace winner |
|-----------------------|------------------|--------------|
| Tournament size       | {2, 3, 4, 5, 7}  | 7            |
| Crossover prob. $p_c$ | [0.5, 0.95]      | 0.77         |
| Tree-size cap         | {20, 30, 40, 60} | 30           |
| Elitism               | {1, 2, 4}        | 2            |

get of 200 experiments, each one a full evolution (population 100, 50 generations) on the training instances, scored by the resulting rule’s RE on the development set. Table 3 lists the parameter space and the winning configuration, which is the one used for every experiment reported in this paper (Table 2). The elite configurations agree on a high selection pressure (tournament size 7) and a crossover probability in the range 0.72–0.77; the tree-size cap does not discriminate, with surviving elites spanning caps of 20, 30 and 40.

## 6. Results

### 6.1. Evolution behaviour and seed stability

Over 30 independent evolutions, training RE converges to  $14.9 \pm 1.5$  (range 12.4–19.7), with most of the improvement realized by generation 25–30 (Figure 2). The evolved rules combine the classical dispatching ingredients in recognizable form. Figure 3 shows the featured rule, obtained with seed 1: 26 nodes over 12 leaves. Every terminal of the tree is non-negative by construction and processing times are at least one time unit, so both guards it contains resolve to the same branch at every decision, and the tree is exactly

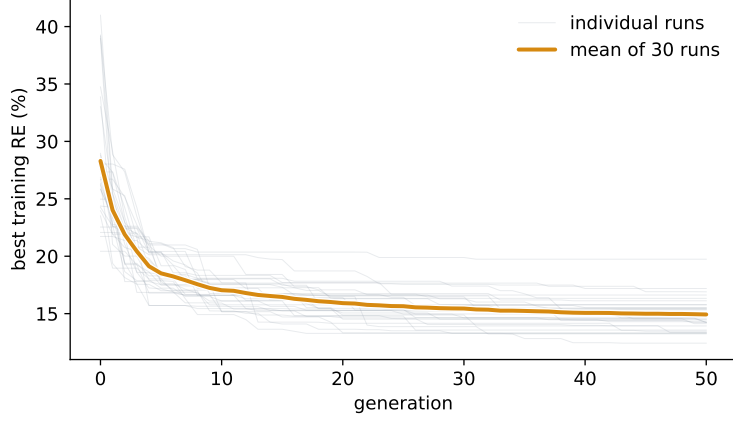


Figure 2: Best training fitness per generation over the 30 independent evolutions: each individual run and their mean.

equivalent on the problem domain to

$$SLACK^2 + 2PT - WKR - WKRW - 1, \quad (12)$$

an identity verified at each of the 37,250 dispatching decisions the rule takes on the benchmark (1.29 million candidate evaluations). Since the lowest value is dispatched, the rule reads directly:  $-WKR$  is most-work-remaining,  $2PT$  is shortest-processing-time at double weight,  $SLACK^2$  penalises quadratically any operation that cannot start at the earliest available time, and the constant is immaterial to the ranking. The remaining term,  $-WKRW$ , is the interval-aware one: among jobs with comparable remaining work the rule advances those whose remaining work is *more uncertain*, resolving that uncertainty early rather than carrying it to the end of the schedule. The evolved rule is thus a weighted blend of MWKR and SPT, regularised by a quadratic earliest-start term and tilted by the uncertainty of the work ahead.

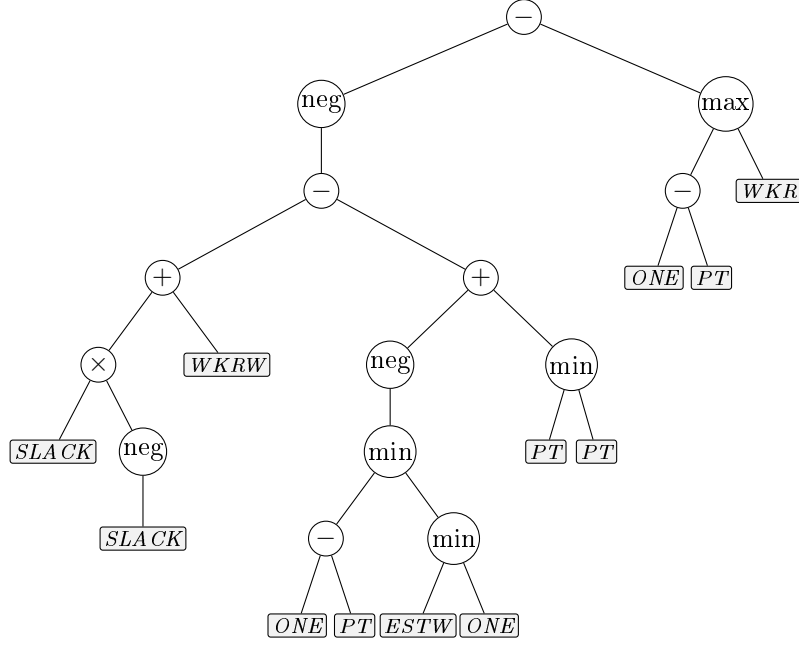


Figure 3: The featured rule (seed 1), which attains 17.71% RE over the 70 instances. Circles are functions and shaded boxes the interval-aware terminals of Table 1; the operation with the lowest value of the expression is dispatched. The tree is shown exactly as evolved; Eq. (12) gives its simplified form.

## 6.2. Generalization to the full benchmark

Table 4 reports single-rollout RE per size class over all 70 instances, as mean  $\pm$  standard deviation across the 30 evolved rules and for the single best rule. The best rule attains 17.71% global mean—evolved only on  $20 \times 15$ , its best class is the unseen  $50 \times 15$  (13.8%), and the same pattern holds on average ( $13.7 \pm 1.5$ ). Because all terminals are per-operation attributes with no size-bound quantities, transfer requires no retraining or rescaling.

Table 4: RE (%) per size class, single deterministic rollout, over all 70 Taillard interval instances: mean  $\pm$  sd across the 30 independently evolved rules, and the best evolved rule.

|            | 15 $\times$ 15 | 20 $\times$ 15 | 20 $\times$ 20 | 30 $\times$ 15 | 30 $\times$ 20 | 50 $\times$ 15 | 50 $\times$ 20 | All              |
|------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|------------------|
| Mean of 30 | 18.1 $\pm$ 1.3 | 18.0 $\pm$ 1.5 | 20.9 $\pm$ 1.7 | 21.4 $\pm$ 1.7 | 25.7 $\pm$ 1.9 | 13.7 $\pm$ 1.5 | 15.1 $\pm$ 1.5 | 18.99 $\pm$ 1.33 |
| Best rule  | 15.7           | 16.6           | 18.1           | 21.1           | 24.1           | <b>13.8</b>    | 14.6           | <b>17.71</b>     |

### 6.3. Position among constructive methods

Table 5 reports every constructive baseline under the protocol of this paper—one deterministic pass per instance, the same decoder and evaluator for all methods—together with a uniformly random dispatcher as a reference point.

Two observations are needed to read the table correctly. First, the plain rules that ignore machine availability (SPT, LPT and the critical ratio CR, the latter being a due-date rule applied here without due dates) perform *worse than random* in this setting. This is a property of the schedule generation scheme, not a defect of the criteria: because a solution is decoded as a permutation whose operations are appended to their machines, systematically dispatching by processing time alone produces highly unbalanced permutations. The same criterion embedded in the Giffler–Thompson conflict set—which restores classical dispatching semantics—behaves as expected (SPT: 627% as a plain rule versus 70.6% under G&T). Second, the criteria that do account for the state of the shop (EST, MOR, MWKR) are clearly better than random, and the strongest baseline overall is G&T-MWKR at 29.5%, which is the reference we use throughout.

Against that strongest baseline, the evolved rules average  $18.99 \pm 1.33$  RE over the 30 evolutions, and the best rule attains 17.71%.

The gap holds at the level of individual instances: the best evolved rule

beats each of the eight deterministic baselines on every one of the 70 instances, and a paired Wilcoxon signed-rank test rejects equality against each of them at  $p < 10^{-6}$  ( $z = -7.3$ ). The narrowest margin over the whole comparison is 3.9 points, against G&T-MWKR on a single  $20 \times 15$  instance; against the best plain rule (EST) the smallest margin is 6.3 points. Per-instance figures are given in Table A.9 (appendix). This quality comes at no evaluation penalty over hand-crafted rules: evaluating the evolved expression tree over the eligible set costs what evaluating a hand-crafted criterion costs. Averaged over the 70 instances, a complete constructive pass takes 335 ms with the evolved rule against 334 ms with MOR and 382 ms with G&T-MWKR, a spread of 14% between the three; all of them reduce to one vectorized pass over the eligible operations per decision, so the choice of priority expression contributes little to the cost of a pass. Both cost and quality thus place the evolved rule with the constructive methods, while closing much of the quality gap that separated them from search-based methods; that comparison is deferred to Section 6.4, where published results are available on a common benchmark.

#### 6.4. Cross-family generalization: the 12 classical instances

The 12 classical instances serve a double purpose: they test the evolved rule on instance families and sizes it has never seen—the rule was evolved on four interval Taillard  $20 \times 15$  instances and is applied here without any re-evolution—and they enable a direct comparison with the *published* results of the IJSP metaheuristics on their own benchmark and metric (Table 6). Those methods belong to different algorithmic families—evolutionary computation and swarm intelligence: a genetic algorithm evolving a population of permutations [1] and three artificial bee colony variants [2, 3], each of

which runs a separate search on every instance. Alongside the deterministic pass, Table 6 reports the randomized extension of Section 4.3 at best-of-64 and best-of-1024:  $N$  independent  $\varepsilon$ -randomized passes of the same rule, of which the best schedule is kept, with no refinement or recombination among samples.

The two families also differ in how they spend computation: best-of-1024 evaluates 1024 mutually independent samples per instance, whereas the metaheuristics evolve a population of 250 individuals [1, 2] through successive generations, each of which depends on the previous one. In wall-clock terms, a single constructive pass takes  $\approx 0.1$  s on instances of this size in our Python implementation, against the 0.5–2.2 s reported for the genetic algorithm and the 1.9–6.8 s reported for fEABC on the same instances [2], both implemented in C++. Best-of-1024, however, requires between 0.6 and 2.4 minutes of sequential Python time depending on instance size, and is therefore *not* faster than the published metaheuristics.

We report this transfer as a property of the method rather than of one rule. Over the 30 independently evolved rules the deterministic pass averages  $19.04 \pm 1.97\%$  RE on these instances, individual rules ranging from 15.3% to 23.3%, so the figure attached to any single rule should be read with that spread in mind. The two benchmarks are referred to different bounds—Taillard’s for the Taillard-derived instances, best-known ones for the classical set—so their RE values are not directly comparable; what does carry across is the margin over the baselines evaluated on the same instances against the same bound. That margin is preserved: on the classical set the evolved rules average 2.40 times better than MOR and 1.58 times better than G&T-MWKR, against 2.40 and 1.55 on the interval Taillard benchmark they were evolved for.

In terms of quality, the featured rule wrapped in GP- $\varepsilon$  at best-of-1024 reaches 11.3% RE against the 9.8% published average of the genetic algorithm, beating it on 3 of the 12 instances; a Wilcoxon signed-rank test on the paired per-instance values puts the difference at the edge of significance ( $z = 1.96$ ,  $p \approx 0.05$ ). The evolved rules therefore do not match the per-instance metaheuristics on this benchmark, and the purpose-built ABC variants remain clearly ahead (5.5–6.4%), as expected of iterative search. What the comparison locates is the constructive frontier: rules evolved once on another benchmark and applied here with no retraining land 1.5 points of RE from a population-based method that searches every instance separately, and well clear of the deterministic rules of Table 5.

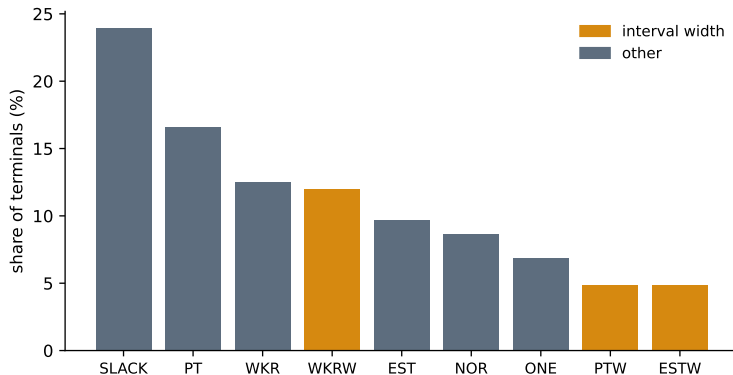


Figure 4: Terminal usage across the 30 evolved rules, as a share of all terminals. Interval-width terminals (amber) expose the magnitude of the uncertainty to the rule.

## 7. Analysis

Beyond aggregate quality, this section examines *why* the evolved rules work: what they are made of, whether the interval-specific ingredients con-

tribute, and how robust the resulting schedules are under executional uncertainty.

### 7.1. Rule anatomy and interpretability

A practical appeal of GP over neural policies is that its output is a readable expression. Evolved rules are compact—mean tree size 26 nodes (20–30), mean depth 9—so a rule can be inspected and simplified by hand. Figure 4 reports how often each terminal is used across the 30 evolved rules. The backbone is the classical scheduling triad—slack (*SLACK*), work remaining (*WKR*) and earliest start (*EST*)—together with worst-case processing time (*PT*), indicating that the evolution rediscovers and refines known dispatching logic rather than exploiting benchmark artefacts. The *interval-width* terminals (*PTW*, *ESTW*, *WKRW*) account for  $\approx 21\%$  of all terminal usage and appear in 27 of the 30 evolved rules: the evolution consistently reaches for uncertainty information, even though worst-case bounds carry the larger share of the signal. Whether that width information translates into better schedules is quantified by the ablation of Section 7.2 and the robustness analysis of Section 7.3.

### 7.2. The value of interval-awareness: terminal ablation

The interval-width terminals are the distinctive ingredient of this setting, so we isolate their contribution by ablation: we repeat the 30 evolutions with and without the width terminals (*PTW*, *ESTW*, *WKRW*) in the terminal set, everything else held fixed, and compare the resulting rules with a paired Wilcoxon signed-rank test on the shared seeds. Table 7 collects the outcome.

Under the makespan objective, the width terminals do not improve the quantity being optimised. Removing the three of them leaves the expected

makespan unchanged:  $18.66 \pm 1.08$  mean RE over the 70 instances without them versus  $18.99 \pm 1.33$  with them, a difference that is not significant ( $z = -0.83$ ) and, if anything, favours the smaller terminal set. They do, however, leave a mark on the schedules: the relative width of the predicted makespan interval is significantly narrower when the rule can read widths ( $12.41 \pm 0.30$  versus  $12.62 \pm 0.23$ ; paired Wilcoxon  $z = 3.40$ ,  $p < 0.001$ ), even though nothing in this objective rewards narrowness. The evolution does reach for these terminals—they appear in 27 of the 30 rules (Section 7.1)—and what it obtains from them here is a tighter interval at no cost in expected makespan.

The makespan fitness gives the evolution no incentive to trade expected makespan for predictability. To test whether it can make that trade when rewarded for it, we repeat the ablation with a *robust* fitness that explicitly rewards predictability, minimising

$$f_\lambda = \overline{C}_{\max} + \lambda (\overline{C}_{\max} - \underline{C}_{\max}), \quad (13)$$

that is, a low worst case *and* a narrow interval, with the weight  $\lambda$  setting the balance between the two; we take  $\lambda = 1$  here. Under this objective the separation widens. Rules evolved with the width terminals produce schedules whose makespan interval is significantly narrower than those evolved without them ( $12.04 \pm 0.75$  versus  $12.47 \pm 0.20$  relative width; paired Wilcoxon  $z = 3.22$ ,  $p < 0.01$ ). Deprived of width terminals, the evolution stays at  $\approx 12.5\%$  under both objectives—12.62 under makespan and 12.47 under the robust fitness—whereas with them it moves from 12.41 to 12.04% as soon as narrowness is rewarded, and further still as  $\lambda$  grows. That movement is paid for in expected makespan (19.62 versus 18.42 RE), which is the trade-off the robust objective asks for.

The weight  $\lambda$  of Eq. (13) governs that trade-off, and sweeping it traces

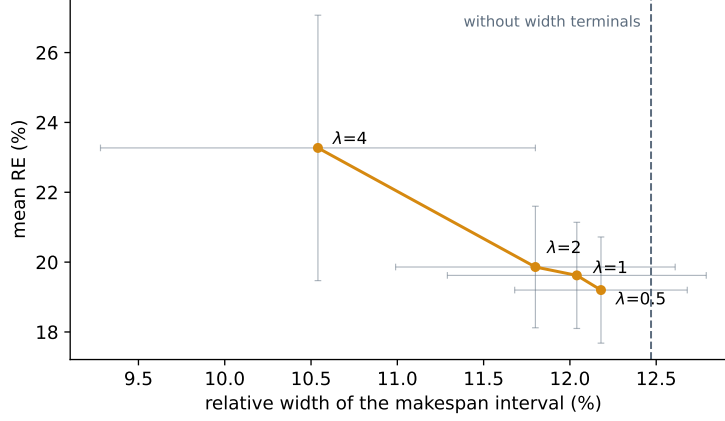


Figure 5: Quality–predictability frontier traced by the weight  $\lambda$  of the robust fitness, Eq. (13). Rules evolved *with* the interval-width terminals; each point is the mean over the independent evolutions of that arm, every rule evaluated on the 70 instances, and the grey bars span  $\pm 1$  sd in both coordinates. The dashed line marks the width reached by the arm evolved *without* them under the robust fitness ( $\lambda=1$ ). Increasing  $\lambda$  buys narrower makespan intervals at the price of a larger expected makespan.

the frontier between quality and predictability (Figure 5). Both quantities move monotonically with  $\lambda$ : as the weight grows from 0.5 to 4, the relative width of the makespan interval falls from  $12.18 \pm 0.50$  to  $10.54 \pm 1.26$  while the expected makespan degrades from  $19.20 \pm 1.52$  to  $23.27 \pm 3.80$  RE. At the far end of the sweep the schedules are therefore narrower than anything the ablated rules can produce, at a cost of about four points of RE. The practitioner can choose an operating point along this curve. No comparable frontier is available to the ablated arm, which stayed at  $\approx 12.5\%$  under both objectives it was evolved for.

Together, these experiments locate *what* interval awareness buys: the width of an interval is actionable information for a dispatching rule, and what it buys is predictability rather than expected makespan. Exposing the

widths leaves the expected makespan unchanged under either objective and narrows the predicted interval under both—modestly when the objective ignores narrowness, substantially when it rewards it, with  $\lambda$  setting how far the trade is taken.

### 7.3. Robustness under executional uncertainty

Relative error measures a-priori quality; the  $\bar{\varepsilon}$  robustness of Eq. (10) measures how well that a-priori prediction survives execution, and is the quantity examined here. We estimate it by Monte Carlo, drawing  $K=1000$  realizations of the durations uniformly on each operation’s interval and executing the fixed processing order on each of them; lower is more robust. To probe behaviour under growing uncertainty we also widen every interval symmetrically about its midpoint by +20% and +40% (clipping negative bounds to zero), re-evaluating the same processing orders.

The evolved rule also produces more robust schedules, on two complementary measures (Table 8). First, its predicted makespan interval is the *tightest*: a relative width of 12.3% against 13.8% (G&T-MWKR) and 14.4% (MOR), so the a-priori uncertainty attached to the schedule is smaller. Second, its executed makespan tracks the prediction most closely:  $\bar{\varepsilon}$  is lower than for both baselines at the nominal width (5.7 versus 6.4 for MOR and 6.3 for G&T-MWKR; paired Wilcoxon over the 70 instances,  $z = -5.96$  and  $z = -4.42$ , both  $p < 0.001$ ), and its advantage *widens* as uncertainty grows (at +40%: 7.6 versus 8.9 and 8.6). Figure 6 shows the per-instance  $\bar{\varepsilon}$  distributions. The figure also includes the ablated arm of Section 7.2: rules evolved *without* the interval-width terminals are as robust as the full rule at every uncertainty level ( $\bar{\varepsilon}$  of 5.7, 6.8 and 7.8 against 5.7, 6.6 and 7.6; the two are statistically indistinguishable at the nominal width,  $z = -0.17$ ), so

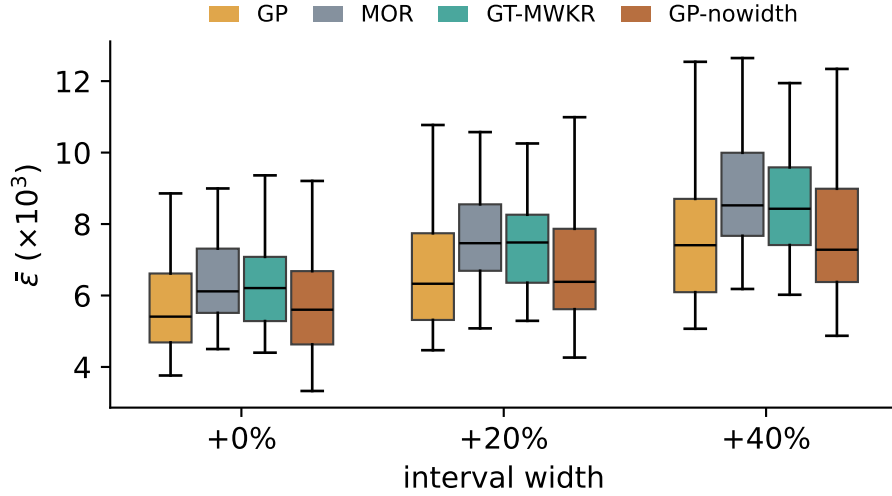


Figure 6: Per-instance  $\bar{\epsilon}$  robustness by method at each uncertainty level (lower is better). The evolved rule’s robustness advantage over the constructive baselines grows with interval width. *GP-nowidth* is the ablated arm of Section 7.2, evolved without the three interval-width terminals: it is indistinguishable from the full rule, which locates the robustness advantage in the quality of the schedules rather than in the rule reading widths.

the advantage does not come from the rule reading interval widths. It comes from producing better schedules in the first place: compact schedules with fewer and shorter critical paths accumulate less uncertainty along them, so both the predicted interval and the realised deviation shrink together.

#### 7.4. Limitations

Three limitations bound the scope of our conclusions. First, all rules operate inside a semi-active SGS; active (insertion-based) generation schemes are known to help weaker constructors, and co-evolving rules with an active SGS in the loop remains untested. Second, the uncertainty model is the  $\pm 15\%$  symmetric interval scheme standard in the IJSP literature; other widths and

asymmetric intervals may change the relative value of uncertainty information. Third, our runtime figures are relative (evolved rules cost the same as hand-crafted ones per pass); absolute times depend on the research-grade Python implementation, but the rankings reported here do not.

## 8. Conclusions and Future Work

We presented what is, to the best of our knowledge, the first GP hyperheuristic for job shop scheduling with interval durations, built on a terminal set that exposes interval widths as first-class attributes, with the evolutionary hyperparameters configured automatically by a 200-experiment irace study. The empirical conclusions are three. (i) Evolved rules dominate every deterministic constructive baseline on the 70-instance benchmark by a wide, statistically significant margin ( $18.99 \pm 1.33$  over 30 evolutions, 17.71% for the best rule, versus 29.5% for the strongest baseline—which the best rule beats on all 70 instances) at the cost of a single constructive pass, and (ii) they generalize zero-shot across all size classes—evolved on  $20 \times 15$ , best on the unseen  $50 \times 15$ —because the representation contains no size-bound quantities. (iii) Exposing the widths of the uncertain quantities does not improve the expected makespan—the worst-case bounds already carry that signal—but it yields significantly narrower makespan intervals, modestly under a makespan objective and substantially under an objective that rewards predictability, with the weight  $\lambda$  setting how far that trade is taken.

Future work follows three lines. First, richer symbolic models: ensembles of complementary rules [22] and co-evolution of construction and tie-breaking. Second, using evolved rules as cheap, high-quality population initializers for the metaheuristics that lead the IJSP: the pools of diverse,

high-quality sequences produced by GP- $\varepsilon$  are directly usable as seeded initial populations, and quantifying the effect of such seeding is ongoing work. Third, a controlled comparison with learned neural constructive policies under matched training and inference budgets—the head-to-head evaluation that the recent literature [21] identifies as missing.

### **CRedit authorship contribution statement**

**Hernán Díaz:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration.

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### **Declaration of competing interest**

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### **Data availability**

The GP hyper-heuristic is implemented from scratch in pure Python/NumPy, with no evolutionary-computation dependencies; evolved rules are stored as plain JSON expression trees and evaluated as drop-in dispatching strategies by the same environment used by every baseline. The

benchmark instances, the generated solutions and rules, and the analysis scripts are shared in a Zenodo repository. [\[TODO: DOI de Zenodo al depositar\]](#)

## Appendix A. Per-instance results

Table A.9 lists, for each of the 70 interval Taillard instances, the crisp lower bound and the single-pass RE (%) of the evolved rule and of the two strongest baselines: EST, the best of the plain dispatching rules, and G&T-MWKR, the best baseline overall.

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Table 5: Constructive baselines on the 70 interval Taillard instances. All the rules are deterministic and build one schedule per instance, so the reported figures are the mean RE (%) over the 70 instances and the standard deviation *across instances*, which measures how uniform a method is over the benchmark rather than any run-to-run variability. The only stochastic entry is the uniformly random dispatcher, included as a reference and averaged over 10 independent dispatches per instance. Rules above the random line ignore machine availability and degrade below it under permutation-based decoding (see text). *ms*: mean wall-clock time of one complete constructive pass, over the same 70 instances and 5 repetitions each, in our research-grade Python implementation; the absolute values are implementation-bound and only the comparison between rows is meaningful, and it was measured for the featured rule rather than for all 30. The two GP entries are the mean over the 30 evolved rules and the best of them; the latter is an order statistic and the former is what a single evolution should be expected to yield.

| Method                   | RE (%)       | sd          | ms        |
|--------------------------|--------------|-------------|-----------|
| LPT                      | 703.6        | 168.0       | 231       |
| SPT                      | 627.2        | 152.2       | 229       |
| CR                       | 625.7        | 140.6       | 256       |
| <i>Random dispatcher</i> | <i>127.2</i> | <i>13.9</i> | <i>85</i> |
| G&T-SPT                  | 70.6         | 10.0        | 295       |
| MWKR                     | 64.7         | 10.4        | 332       |
| MOR                      | 45.5         | 10.1        | 334       |
| EST                      | 42.3         | 10.6        | 279       |
| G&T-MWKR                 | <b>29.5</b>  | 6.4         | 382       |
| GP rule (mean of 30)     | 18.99        | 4.96        | —         |
| GP rule (best of 30)     | <b>17.71</b> | 5.23        | 335       |

Table 6: RE (%) on the 12 classical interval instances: our methods (deterministic single pass and GP- $\varepsilon$  best-of- $N$ ; rule evolved on interval Taillard 20 $\times$ 15, applied zero-shot) versus the published average results (30 runs per instance) of the IJSP metaheuristics GA [1], ABC<sub>E3</sub>, fEABC [2] and ESABC [3]. LB is the best-known lower bound of the expected makespan. Per-instance rows are the single featured rule; the last row gives the mean  $\pm$  sd of the deterministic pass over all 30 independently evolved rules, so that the transfer is read as a property of the method and not of a single rule.

| Inst.      | LB   | Ours (zero-shot) |                        |                          | Published (avg of 30) |                   |       |       |
|------------|------|------------------|------------------------|--------------------------|-----------------------|-------------------|-------|-------|
|            |      | GP               | GP- $\varepsilon_{64}$ | GP- $\varepsilon_{1024}$ | GA                    | ABC <sub>E3</sub> | fEABC | ESABC |
| FT10       | 930  | 20.4             | 8.4                    | 6.6                      | 5.2                   | 4.1               | 3.5   | 3.0   |
| FT20       | 1165 | 13.2             | 12.1                   | 10.8                     | 4.4                   | 1.7               | 1.8   | 1.8   |
| La21       | 1046 | 15.0             | 14.4                   | 9.0                      | 5.0                   | 5.0               | 4.2   | 4.0   |
| La24       | 935  | 10.1             | 8.8                    | 8.6                      | 6.3                   | 5.1               | 4.9   | 5.0   |
| La25       | 977  | 11.8             | 11.6                   | 7.6                      | 5.1                   | 3.9               | 3.4   | 2.7   |
| La27       | 1235 | 25.0             | 13.7                   | 11.3                     | 10.2                  | 4.7               | 4.6   | 4.1   |
| La29       | 1152 | 13.4             | 14.2                   | 11.0                     | 14.2                  | 8.6               | 7.4   | 7.0   |
| La38       | 1196 | 15.0             | 13.3                   | 9.4                      | 9.2                   | 6.9               | 6.1   | 5.8   |
| La40       | 1222 | 13.3             | 12.8                   | 9.9                      | 8.7                   | 4.2               | 4.6   | 4.1   |
| ABZ7       | 656  | 19.7             | 14.3                   | 11.7                     | 12.5                  | 7.3               | 6.6   | 6.7   |
| ABZ8       | 645  | 24.7             | 22.1                   | 17.7                     | 18.5                  | 12.1              | 11.1  | 10.9  |
| ABZ9       | 661  | 33.4             | 25.3                   | 22.1                     | 18.0                  | 13.1              | 11.6  | 11.2  |
| Mean       |      | 17.9             | 14.2                   | 11.3                     | 9.8                   | 6.4               | 5.8   | 5.5   |
| Mean of 30 |      | 19.0 $\pm$ 2.0   | —                      | —                        |                       |                   |       |       |

Table 7: Terminal ablation under two fitness objectives (30 independent evolutions per arm; rank-based tests between arms). *Width* is the relative width of the predicted makespan interval. Interval-width terminals do not help when the objective is the expected makespan—the two arms are then indistinguishable on both measures—but do produce significantly narrower schedules when the objective rewards predictability.

| Fitness objective               | Terminal set   | RE (%)                      | Width (%)                          |
|---------------------------------|----------------|-----------------------------|------------------------------------|
| Makespan $E[\mathbf{C}_{\max}]$ | full           | $18.99 \pm 1.33$            | $12.41 \pm 0.30$                   |
|                                 | without widths | $18.66 \pm 1.08$            | $12.62 \pm 0.23$                   |
|                                 | <i>test</i>    | <i>n.s.</i> ( $z = -0.83$ ) | $z = 3.40, p < 0.001$              |
| Robust, Eq. (13)                | full           | $19.62 \pm 1.52$            | <b><math>12.04 \pm 0.75</math></b> |
|                                 | without widths | $18.42 \pm 0.87$            | $12.47 \pm 0.20$                   |
|                                 | <i>test</i>    | —                           | $z = 3.22, p < 0.01$               |

Table 8: Robustness over the 70 instances (lower is better on both measures). *Rel. width*: relative width of the predicted makespan interval,  $(\overline{C}_{\max} - \underline{C}_{\max})/E[C_{\max}]$  (%).  $\bar{\varepsilon} (\times 10^3)$ : mean deviation of the executed makespan from its prediction, at nominal interval widths and widened by +20% and +40% (Monte Carlo,  $K=1000$  realizations, common random numbers across methods). *Ablated*: the best rule evolved without the interval-width terminals, the arm of Section 7.2. All four methods come from a single evaluation run.

| Method                        | Rel. width (%) | $\bar{\varepsilon} (\times 10^3)$ |            |            |
|-------------------------------|----------------|-----------------------------------|------------|------------|
|                               |                | +0%                               | +20%       | +40%       |
| GP rule (best of 30)          | <b>12.3</b>    | <b>5.7</b>                        | <b>6.6</b> | <b>7.6</b> |
| GP rule, ablated (best of 30) | 12.5           | 5.7                               | 6.8        | 7.8        |
| G&T-MWKR                      | 13.8           | 6.3                               | 7.5        | 8.6        |
| MOR                           | 14.4           | 6.4                               | 7.7        | 8.9        |

Table A.9: Per-instance RE (%) of the evolved rule (GP) and of the two strongest baselines, EST and G&T-MWKR (single deterministic pass). LB is the crisp Taillard lower bound.

| Inst. | LB   | EST  | G&T  | GP   | Inst. | LB   | EST  | G&T  | GP   |
|-------|------|------|------|------|-------|------|------|------|------|
| TA1   | 1231 | 49.9 | 26.1 | 19.3 | TA36  | 1819 | 41.1 | 42.8 | 23.8 |
| TA2   | 1244 | 24.9 | 27.8 | 14.2 | TA37  | 1771 | 43.4 | 35.6 | 18.4 |
| TA3   | 1218 | 33.0 | 33.2 | 14.1 | TA38  | 1673 | 45.0 | 28.0 | 20.1 |
| TA4   | 1175 | 32.0 | 27.7 | 13.0 | TA39  | 1795 | 46.6 | 28.9 | 22.4 |
| TA5   | 1224 | 27.2 | 33.0 | 14.7 | TA40  | 1651 | 46.9 | 42.3 | 26.1 |
| TA6   | 1238 | 26.8 | 18.2 | 13.7 | TA41  | 1906 | 44.6 | 45.0 | 33.4 |
| TA7   | 1227 | 30.7 | 26.1 | 17.7 | TA42  | 1884 | 71.1 | 38.1 | 24.3 |
| TA8   | 1217 | 23.7 | 25.0 | 14.8 | TA43  | 1809 | 68.7 | 39.8 | 22.3 |
| TA9   | 1274 | 37.7 | 28.2 | 20.0 | TA44  | 1948 | 52.1 | 34.7 | 25.8 |
| TA10  | 1241 | 37.0 | 22.1 | 15.4 | TA45  | 1997 | 50.2 | 33.7 | 14.9 |
| TA11  | 1357 | 56.2 | 30.7 | 17.5 | TA46  | 1957 | 60.8 | 31.6 | 22.4 |
| TA12  | 1367 | 48.8 | 34.6 | 17.6 | TA47  | 1807 | 46.8 | 30.1 | 25.6 |
| TA13  | 1342 | 60.2 | 26.9 | 13.3 | TA48  | 1912 | 59.1 | 33.7 | 21.1 |
| TA14  | 1345 | 39.5 | 29.7 | 13.2 | TA49  | 1931 | 51.0 | 35.3 | 20.1 |
| TA15  | 1339 | 48.1 | 30.6 | 16.7 | TA50  | 1833 | 61.4 | 46.2 | 31.3 |
| TA16  | 1360 | 45.2 | 34.0 | 15.6 | TA51  | 2760 | 33.8 | 34.7 | 23.2 |
| TA17  | 1462 | 58.9 | 17.7 | 13.9 | TA52  | 2756 | 38.0 | 26.5 | 12.7 |
| TA18  | 1377 | 43.8 | 31.8 | 23.6 | TA53  | 2717 | 34.1 | 18.3 | 11.8 |
| TA19  | 1332 | 45.8 | 29.3 | 20.0 | TA54  | 2839 | 20.9 | 16.1 | 8.2  |
| TA20  | 1348 | 45.7 | 24.1 | 15.1 | TA55  | 2679 | 31.7 | 29.2 | 15.5 |
| TA21  | 1642 | 33.4 | 28.0 | 20.2 | TA56  | 2781 | 34.8 | 19.8 | 10.9 |
| TA22  | 1561 | 44.2 | 30.8 | 17.4 | TA57  | 2943 | 32.9 | 20.0 | 9.6  |
| TA23  | 1518 | 42.8 | 35.3 | 17.8 | TA58  | 2885 | 37.4 | 26.3 | 13.6 |
| TA24  | 1644 | 56.3 | 23.8 | 14.1 | TA59  | 2655 | 26.6 | 28.2 | 16.7 |
| TA25  | 1558 | 45.8 | 33.8 | 16.6 | TA60  | 2723 | 41.7 | 19.8 | 15.5 |
| TA26  | 1591 | 42.8 | 31.2 | 26.9 | TA61  | 2868 | 45.7 | 25.4 | 15.4 |
| TA27  | 1652 | 50.0 | 31.1 | 21.4 | TA62  | 2869 | 43.0 | 27.6 | 17.2 |
| TA28  | 1603 | 36.8 | 26.8 | 12.4 | TA63  | 2755 | 38.8 | 23.8 | 13.4 |
| TA29  | 1583 | 28.8 | 32.0 | 17.2 | TA64  | 2702 | 38.2 | 28.4 | 12.2 |
| TA30  | 1528 | 34.2 | 36.1 | 17.0 | TA65  | 2725 | 42.9 | 30.0 | 20.5 |
| TA31  | 1764 | 30.7 | 34.4 | 24.5 | TA66  | 2845 | 41.8 | 28.0 | 11.9 |
| TA32  | 1774 | 42.7 | 33.3 | 19.0 | TA67  | 2825 | 31.7 | 26.1 | 15.2 |
| TA33  | 1788 | 51.3 | 34.8 | 27.5 | TA68  | 2784 | 50.1 | 19.1 | 10.6 |
| TA34  | 1828 | 45.9 | 31.8 | 20.7 | TA69  | 3071 | 35.3 | 22.8 | 12.9 |
| TA35  | 2007 | 30.3 | 24.1 | 8.4  | TA70  | 2995 | 42.9 | 23.4 | 16.7 |